

28 years of CodeWars archives can create quite a cleanup job for the bots in the basement.

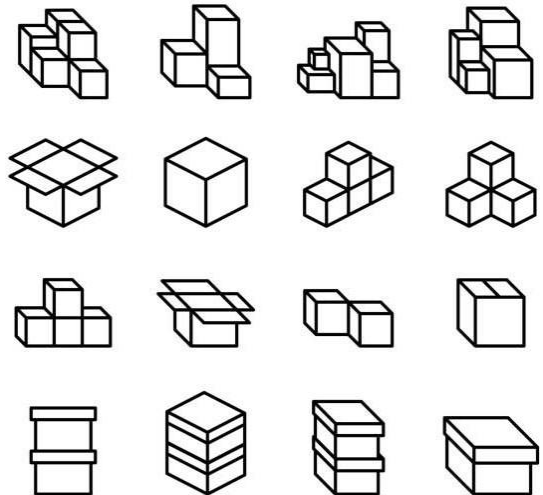
Each room in the basement has collected its own pile of data. Each day we drop the bot at a different location and tell it to find the basement's exit room in the shortest time, shredding boxes of old data as it goes.

The amount of data in each room directly affects how long it takes to enter the room. The bot knows the amount of clutter for each room, and has very basic instructions on how to pick a path with the shortest time. For each map provided, find its path.

Input

The first line of input is two integers separated by a space, the height (H, max 10) and width (W, max 20) of the room. Each of the next H lines has W characters showing the map.

Each room is marked, with '*' for the START, '@' for the FINISH, '#' for a wall (a blocked room), or a label for the number of seconds it takes to clear and enter the room: 1-9,A-Z. A=10, Z=35. It takes 1 second to enter the FINISH room.



```
6 10      4 5
1111441111 11111
1P112#1321 12224
2292121811 *353@
22#1##T211 12341
23*Z@11311
111Y111111
```

Output

Print the time of the fastest path in seconds, then reprint the solved grid showing only start '*', finish '@', walls '#' and the solution path. All other squares should be replaced with a period '.'.

Remember to include 1 second for entering the final square. At every intersection where multiple paths have the same time, prioritize Down before Right before Up before Left. For example, if paths Right and Up have the same total time to the finish, choose Right. This may lead to a longer distance, but it's the same time, and distance doesn't matter.

```
36 seconds      11 seconds
111.....      .....
1.11.#132.      1222.
2..2121.1.      *.3@
2.#.##..1.      .....
3.*.@11.1.
111...111.
```

Discussion

In the first example above, moving only 2 squares to the right is also a time of 36 seconds (Z+1). Since they're the same time, the bot prioritizes the path going Down over the path going Right.

In the second example, the path through the top '1' rooms is also 11 seconds, but Right is prioritized before Up, so the bot makes a right turn to the 2.